

# On the Diffusion Geometry of Stochastic Gradient Descent: Spectral and Stochastic Analysis for Supervised Image Classification

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**Abstract**—Stochastic Gradient Descent (SGD) remains the dominant optimization method in modern deep learning, yet its implicit regularization behavior and stochastic dynamics are still not fully understood. This paper presents a diffusion-geometric framework for analyzing SGD through stochastic differential equations (SDEs), Hessian spectral analysis, and partial differential equation-informed modeling. The proposed framework models SGD trajectories as continuous stochastic processes and investigates how gradient noise covariance, parameter diffusion, and Hessian eigenvalue evolution influence optimization behavior and generalization performance. Experiments conducted on supervised image classification tasks using convolutional neural networks trained on MNIST demonstrate that SGD exhibits highly anisotropic diffusion patterns correlated with curvature properties of the loss landscape. Furthermore, the empirical results validate the effectiveness of SDE approximations for describing SGD dynamics over finite training horizons.

**Index Terms**—Stochastic Gradient Descent, diffusion geometry, stochastic differential equations, Hessian spectrum, Langevin dynamics, optimization

## I. INTRODUCTION

Stochastic Gradient Descent (SGD) serves as the foundational optimization algorithm for modern machine learning systems, particularly deep neural networks. Despite its simplicity, SGD consistently achieves strong empirical performance across high-dimensional optimization problems. Traditional analyses of SGD focus primarily on convergence rates and optimization guarantees; however, recent research suggests that the stochasticity of SGD plays a significant role in generalization, implicit regularization, and exploration of the loss landscape.

Recent theoretical advances have shown that SGD can be approximated as a stochastic differential equation under suitable scaling assumptions. This interpretation provides a mathematical bridge between optimization theory, statistical physics, and diffusion processes. By viewing SGD as a stochastic dynamical system, it becomes possible to analyze the geometry of optimization trajectories, noise covariance structures, and spectral properties of the Hessian matrix.

This work introduces a computational and theoretical framework for studying the diffusion geometry of SGD in supervised image classification systems.

The primary contributions of this paper are:

- A continuous-time SDE formulation of SGD trajectories.
- A spectral framework for tracking Hessian eigenvalue evolution during training.

- Empirical characterization of anisotropic gradient noise.
- Integration of Langevin dynamics and Fokker–Planck modeling for optimization analysis.
- A reproducible software framework for studying optimization geometry in deep learning.

## II. RELATED WORK

The stochastic interpretation of SGD has been extensively studied in recent years. Mandt et al. proposed interpreting SGD as approximate Bayesian inference using Ornstein–Uhlenbeck dynamics, establishing a relationship between stochastic optimization and probabilistic inference.

Recent literature further demonstrated that SGD preferentially explores flatter minima due to anisotropic gradient noise. Hessian spectral analysis has also emerged as an important tool for understanding optimization dynamics in large-scale neural networks.

## III. THEORETICAL FRAMEWORK

### A. SGD as a Stochastic Differential Equation

Consider the standard SGD update rule:

$$\theta_{k+1} = \theta_k - \eta \nabla \hat{L}(\theta_k) \quad (1)$$

where  $\eta$  denotes the learning rate and  $\nabla \hat{L}(\theta_k)$  is the stochastic gradient estimate.

Under continuous-time scaling assumptions, SGD can be approximated by:

$$d\theta_t = -\nabla L(\theta_t)dt + \sqrt{2\gamma B(\theta_t)}dW_t \quad (2)$$

where:

- $L(\theta)$  is the empirical loss function,
- $B(\theta)$  is the gradient noise covariance tensor,
- $\gamma$  is the diffusion scaling parameter,
- $W_t$  is a Wiener process.

### B. Fokker–Planck Dynamics

The probability density associated with SGD evolution follows the Fokker–Planck equation:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (\rho \nabla L) + \gamma \nabla \cdot (\nabla \cdot (B\rho)) \quad (3)$$

### C. Hessian Spectral Analysis

The Hessian matrix provides local curvature information about the loss landscape:

$$H(\theta) = \nabla^2 L(\theta) \quad (4)$$

Tracking the Hessian eigenvalue spectrum throughout training allows characterization of sharp and flat regions encountered by SGD.

## IV. METHODOLOGY

### A. System Architecture

The proposed framework is organized into four primary components:

- 1) Gradient noise estimation
- 2) Hessian spectral computation
- 3) Diffusion tensor analysis
- 4) Stochastic process simulation

### B. Dataset and Model

Experiments were conducted using the MNIST handwritten digit dataset. A convolutional neural network architecture was implemented using PyTorch. SGD optimization was performed with configurable batch sizes, learning rates, and momentum parameters.

### C. Gradient Noise Estimation

Gradient covariance matrices were computed across mini-batches to estimate the stochastic structure of SGD updates. Principal Component Analysis (PCA) was subsequently applied to identify dominant diffusion directions.

### D. Hessian Spectrum Computation

Hessian eigenvalues were estimated during training to analyze curvature evolution. Spectral trajectories were logged across optimization iterations.

### E. Diffusion Simulation

Continuous-time stochastic simulations were performed using Langevin dynamics and Ornstein–Uhlenbeck approximations to compare theoretical predictions with empirical SGD trajectories.

## V. EXPERIMENTAL RESULTS

### A. Gradient Noise Geometry

The experiments revealed that SGD noise exhibits strong anisotropic behavior. Diffusion intensity varied significantly across parameter directions, suggesting that SGD does not explore parameter space uniformly.

Principal component analysis demonstrated that a small subset of eigen-directions dominated the diffusion process.

### B. Hessian Spectrum Evolution

The Hessian eigenvalue distribution evolved dynamically during training. Early training phases exhibited rapidly increasing dominant eigenvalues corresponding to sharp curvature regions. Later training phases showed spectral stabilization and reduced curvature magnitudes.

### C. SDE Approximation Accuracy

Continuous-time stochastic simulations closely approximated empirical SGD trajectories over moderate time horizons.

### D. Implicit Regularization Effects

The diffusion structure induced by stochastic gradients produced implicit regularization behavior analogous to explicit smoothing penalties.

## VI. DISCUSSION

The results demonstrate that diffusion geometry provides a useful framework for understanding optimization in deep learning systems. The stochasticity introduced by SGD contributes meaningfully to trajectory shaping, exploration behavior, and implicit regularization.

The observed relationship between Hessian curvature and gradient covariance supports emerging theories connecting optimization geometry with generalization performance.

## VII. LIMITATIONS AND FUTURE WORK

This work currently focuses on supervised image classification using MNIST-scale datasets and compact CNN architectures. Future work may include:

- Scaling analysis to transformer architectures
- Extending spectral analysis to adaptive optimizers
- Investigating diffusion geometry in reinforcement learning systems
- Applying manifold learning techniques to optimization trajectories

## VIII. CONCLUSION

This paper introduced a diffusion-geometric framework for studying SGD through stochastic differential equations, Hessian spectral analysis, and PDE-informed modeling. By modeling SGD as a stochastic dynamical system, the framework provides deeper insight into optimization geometry, implicit regularization, and training dynamics.

Experimental results demonstrated strong relationships between gradient noise covariance, Hessian curvature, and parameter diffusion behavior.

## REFERENCES

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